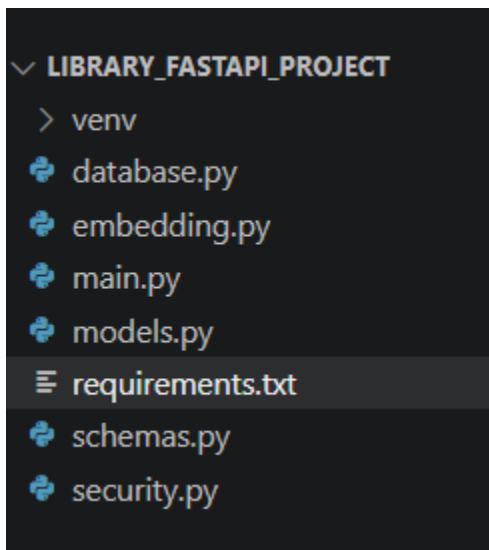


PRACTICAL : WEEK 5

1. File Creation



2. Install packages

```
Downloading cffi-2.1.1-cp311-cp311-win_amd64.whl (185 kB)
185.1/185.1 kB 747.2 kB/s eta 0:00:00
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached idna-3.19-py3-none-any.whl (68 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
48.2/48.2 kB 1.2 MB/s eta 0:00:00
Installing collected packages: passlib, typing-extensions, six, python-multipart, pycparser, pyasn1, idna, h11, greenlet, click, bcrypt,
yncmy, annotated-types, annotated-doc, uvicorn, typing-inspection, sqlalchemy, rsa, pydantic-core, ecdsa, cffi, anyio, starlette, python-
se, pydantic, cryptography, fastapi
Successfully installed annotated-doc-0.0.5 annotated-types-0.8.0 anyio-4.15.1 asyncmy-0.2.14 bcrypt-5.0.0 cffi-2.1.1 click-8.5.0 cryptogr
hy-50.0.1 ecdsa-0.19.2 fastapi-0.141.1 greenlet-3.5.6 h11-0.16.0 idna-3.19 passlib-1.7.4 pyasn1-0.6.4 pycparser-3.0 pydantic-2.13.5 pydan
c-core-2.46.5 python-jose-3.5.0 python-multipart-0.0.32 rsa-4.9.1 six-1.17.0 sqlalchemy-2.0.53 starlette-1.6.0 typing-extensions-4.16.0 t
ing-inspection-0.4.4 uvicorn-0.53.0

[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\anany\OneDrive\Desktop\library_fastapi_project>
```

3. MySQL setup

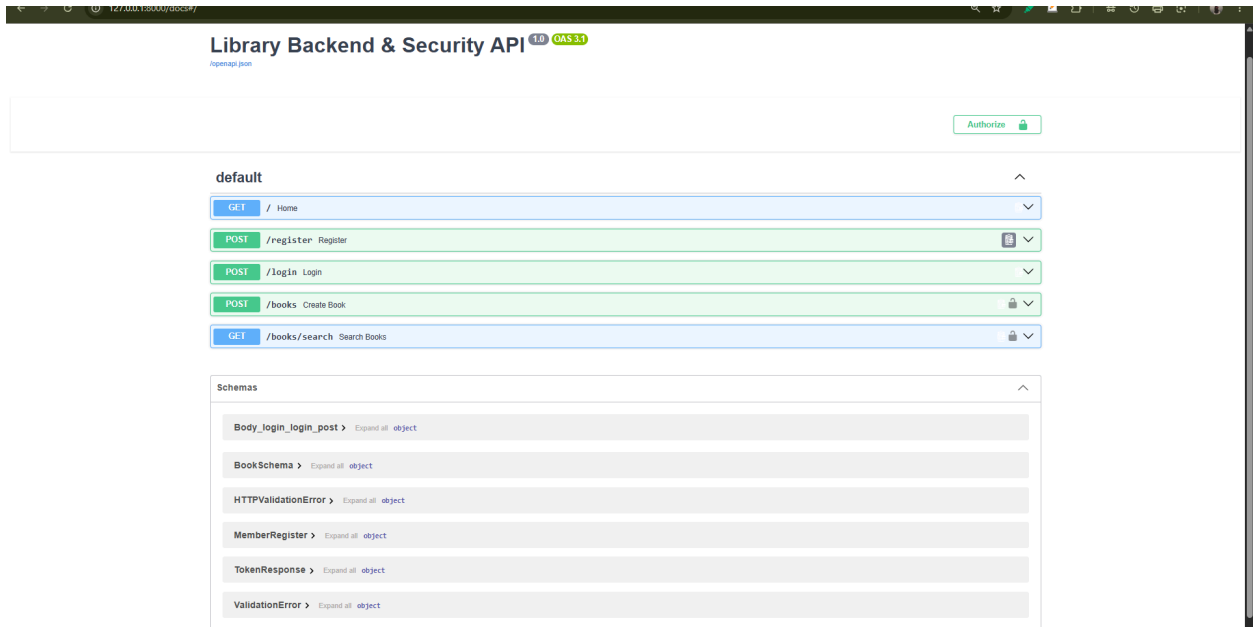
The screenshot displays the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The Navigator pane on the left shows a list of schemas: 2520080069_ananya, bankdb, bookflow_db, kludb, sys, and university. The central Query Editor contains the following SQL code:

```
1 CREATE DATABASE library_db;
2 USE library_db;
3 CREATE TABLE IF NOT EXISTS members (
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     username VARCHAR(100) UNIQUE NOT NULL,
6     password_hash VARCHAR(255) NOT NULL,
7     role VARCHAR(30) NOT NULL DEFAULT 'member'
8 );
9 CREATE TABLE IF NOT EXISTS books (
10    id INT AUTO_INCREMENT PRIMARY KEY,
11    title VARCHAR(250) NOT NULL,
12    author VARCHAR(200) NOT NULL,
13    isbn VARCHAR(13) UNIQUE NOT NULL,
14    price DECIMAL(10,2) NOT NULL,
15    published_date DATE NOT NULL,
16    description TEXT,
17    embedding JSON
18 );
```

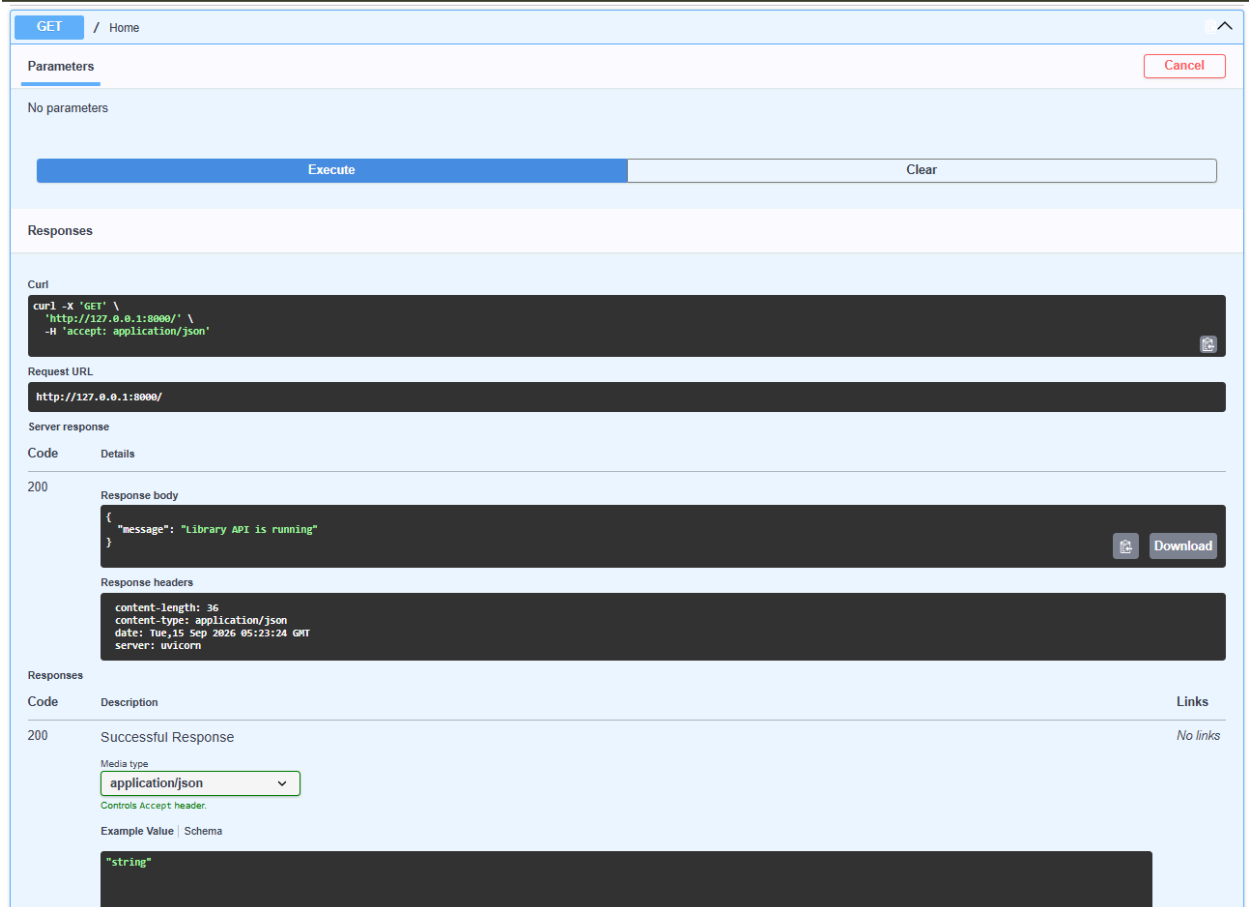
The Output pane at the bottom shows the execution results of the SQL statements:

#	Time	Action	Message
1	10:36:17	CREATE DATABASE library_db	1 row(s) affected
2	10:39:49	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(100) UNIQUE NOT NULL, password...	Error Code: 1046. N
3	10:40:01	USE library_db	0 row(s) affected
4	10:40:05	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(100) UNIQUE NOT NULL, password...	0 row(s) affected
5	10:40:08	CREATE TABLE IF NOT EXISTS books (id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(250) NOT NULL, author VARCHAR(200) NO...	0 row(s) affected

4. API Testing



GET :



POST : Register

POST

/register Register

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "username": "ananya",
  "password": "library123"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/register' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "username": "ananya",
    "password": "library123"
  }'
```

Request URL

```
http://127.0.0.1:8080/register
```

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/register' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "username": "ananya",
    "password": "library123"
  }'
```

Request URL

```
http://127.0.0.1:8080/register
```

Server response

Code

Details

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "ananya",
  "role": "member"
}
```

Download

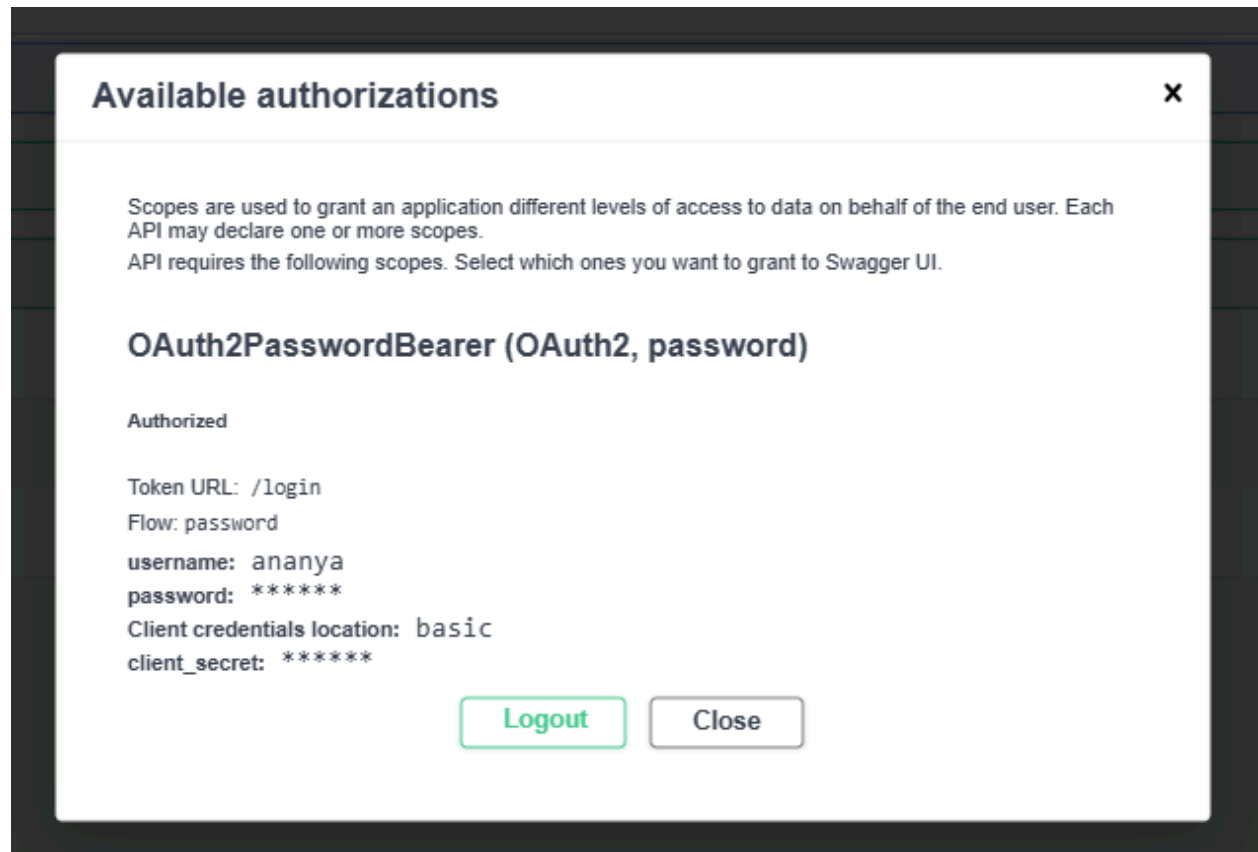
Response headers

```
content-length: 80
content-type: application/json
date: Tue, 15 Sep 2020 05:27:26 GMT
server: indom
```

Responses

Code	Description	Links
200	Successful Response	No links
	Media type application/json Content Accept header Example Value Schema "string"	
422	Validation Error	No links
	Media type application/json Example Value Schema <pre>{ "detail": { "loc": ["string"] } }</pre>	

Authorization



POST : Login

POST /login Login

Parameters

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)

pattern: "passwords"

password

☐ Send empty value

username * required

string

ananya

☐ Send empty value

password * required

string(\$password)

☐ Send empty value

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

string

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Execute

Clear

Responses

Curl

curl -X 'POST' \n\n'http://127.0.0.1:8000/login' \n-H 'accept: application/json' \n-H 'content-type: application/x-www-form-urlencoded' \n-d 'grant_type=password&username=ananya&password=library123&scope=&client_id=string&client_secret=*****'

Request URL

http://127.0.0.1:8000/login

Server response

Code

Details

200

Response body

{\n "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhbmFueWVlIiwiaWF0IjoxNjY1MjI0MDY.8KLEP-CjI7xf42IM-tqMTA1bWt8R6b3hpjM2ID0Y",\n "token_type": "bearer"\n}

Response headers

content-length: 188\ncontent-type: application/json\ndate: Tue, 15 Sep 2026 05:30:32 GMT\nserver: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links
	Media type application/json Controls Accept header Example Value Schema {\n "access_token": "string",\n "token_type": "string"\n}	
422	Validation Error	No links

POST: Books

POST

/books

Create Book

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

Edit Value

Schema

{
 "title": "Python for Beginners",
 "author": "John Smith",
 "isbn": "9781234567890",
 "price": 450.00,
 "published_date": "2024-06-10",
 "description": "An introduction to Python Programming"
}

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \  
  'http://127.0.0.1:8000/books' \  
  -H 'accept: application/json' \  
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhbmFuei1Cjy62x1IjoibWVtYmwy11ki2ZhdWtJc29wZS50XDUzZC9fQ.0eZ1bm85S3p1bHc4S6a2bTdyCIRfS1d7OUg4S2801' \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "title": "Python for Beginners",  
    "author": "John Smith",  
    "isbn": "9781234567890",  
    "price": 450.00,  
    "published_date": "2024-06-10",  
    "description": "An introduction to Python Programming"  
  }'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

201

Response body

{
 "message": "Book added successfully",
 "book_id": 1,
 "added_by": "ananya"
}

Download

Response headers

content-length: 69
content-type: application/json
date: Tue, 15 Sep 2026 05:33:07 GMT
server: uvicorn

Responses

Code

Description

Links

201

Successful Response

No links

Media type

application/json

Controls Accept header:

Example Value | Schema

"string"

422

Validation Error

No links

GET: 422 Error

GET

/books/search

Search Books

Parameters

Cancel

Name	Description
q required	
string	978123
(query)	minLength: 1
limit	
integer	5
(query)	maximum: 20 minimum: 1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/books/search?q=978123&limit=5' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhbmVudWVyeVJlZDZhdWljZXNzZg5MDUzZnc0Q0.0eZ1bnR5I3cp1BHK4sGa2bsTdyC1Rf51dZDUjg92B01'
```

Request URL

```
http://127.0.0.1:8080/books/search?q=978123&limit=5
```

Server response

Code	Details
200	Response body <pre>{ "query": "978123", "results": [{ "id": 1, "title": "Python for Beginners", "author": "John Smith", "isbn": "9781234567890", "price": 450, "published_date": "2024-06-10", "description": "An introduction to Python Programming", "similarity": -0.10866565874486478 }] }</pre>

Responses		
Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"	No links
422	Validation Error Media type application/json Example Value Schema { "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }	No links

Non- authorized User - GET / books

GET

/books/search

Search Books

Parameters

Cancel

Name	Description
q required string (query)	9781234567890 minLength: 1
limit integer (query)	5 maximum: 20 minimum: 1

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=9781234567890&limit=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=9781234567890&limit=5
```

Server response

Code	Details
401 Undocumented	Error: Unauthorized

Response body

```
{
  "detail": "Not authenticated"
}
```

 Download

Response headers

```
content-length: 30
content-type: application/json
date: Tue, 15 Sep 2026 05:36:54 GMT
server: uvicorn
www-authenticate: Bearer
```

Responses